

Ziyu Peng

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Education

University of Colorado Boulder

2026–Present

Undergraduate Student, Statistics and Data Science

Xi'an Jiaotong-Liverpool University (XJTLU)

2025–2026

First-Year Undergraduate Study, *Data Science and Big Data Technology*

School of AI and Advanced Computing, XJTLU Entrepreneur College (Taicang)

GPA: 3.88/4.00 | Calculus I–III, Linear Algebra

Research Interests

Mechanistic interpretability; reliable LLM/VLM reasoning; uncertainty-aware inference, calibration, and selective prediction.

Submitted Manuscript

Scale Is Not Symmetric: Directional Scale Reliability in Chart Question Answering

2026

Multimodal inference reliability; chart question answering; policy selection

- Built **ScaleQA-Natural**, a 520-example audited diagnostic set for decimal/percent and unit-scale conflicts in chart QA.
- Evaluated prompting, directional LoRA adapters, verifiers, calculators, likelihood scoring, and policy selectors to separate candidate generation from candidate selection.
- A targeted adapter improved held-out ChartQA accuracy from **88.60% to 89.32%** and recovered **49.52%** of strict scale errors; broader diagnostics identified regime selection as the main remaining bottleneck.

Selected Research Projects

Ambiguity Is Not One Scalar: Explicit Referents Improve VQA Routing

2026

Conditional multimodal inference; selective prediction; calibration

- Designed a two-stage VQA router over **ANSWER / CLARIFY / ABSTAIN**, using explicit referent enumeration for uncertain or focus-sensitive inputs.
- On a 26,425-example diagnostic benchmark, referent enumeration recovered **20/22** low-score focus-ambiguity false negatives, versus **0/22** for answer self-consistency.

PW-OFPC: Precision-Weighted Object-File Predictive Coding

2025–2026

Uncertainty-aware inference; object-centric reasoning; candidate comparison

- Built an uncertainty-aware abstract visual reasoning pipeline that preserves object-file posteriors and compares answer candidates jointly using predictive-coding residuals.
- On public RAVEN, the inference ladder improved macro accuracy from **0.537 to 0.857**; controlled experiments separated object-formation failures from residual and candidate-selection failures.

Honors & Awards

- CU Boulder Transfer Excellence Scholarship, 2026** – \$12,000 (\$6,000/year), awarded to academically promising nonresident transfer students.
- XJTLU Academic Scholarship (University Academic Achievement Award), 2026/27** – awarded for academic performance in 2025–26.
- Third Prize, XJTLU Biostatistics Competition, 2026** – only first-year student among prize recipients.

Technical Skills

Programming

Python, R

Tools

Linux, Git/GitHub, LaTeX

Machine Learning

PyTorch, Transformers, LLMs/VLMs, LoRA

Methods

Calibration, selective prediction, inference-time routing, verification